

Abstract

Lightweight authenticated encryption has become essential for modern embedded and IoT systems, where security must be achieved under strict constraints on area, power, and computational resources. Schwaemm128-128, a finalist in the NIST Lightweight Cryptography standardization process, is based on the SPARKLE permutation, an ARX (Addition–Rotation–XOR) construction optimized for efficiency on small microarchitectures. This project investigates the practical performance of Schwaemm128 on FPGA-implemented RISC-V soft-core processors, focusing on two widely adopted architectures: PicoRV32, known for its minimal footprint, and VexRiscv, which offers pipelining and configurability for higher throughput. We implemented Schwaemm128 entirely in C and integrated hardware cycle counters to measure encryption and decryption latency on both cores. Each soft core was synthesized on FPGA and tested using a bare-metal runtime environment to eliminate operating-system overhead. Benchmarking results show that VexRiscv consistently outperforms PicoRV32 by 4–5×, primarily due to instruction-level parallelism and deeper pipelines. In addition, C implementations achieved approximately 2× higher performance compared to Rust, reflecting differences in compiler optimizations and register allocation on small RISC-V targets. Our findings demonstrate that SPARKLE-based authenticated encryption is practical even on extremely lightweight microarchitectures with limited hardware support, while higher-performance soft cores significantly reduce encryption latency for security-critical applications. This work contributes empirical insight into how ARX-based cryptographic primitives behave on real RISC-V hardware and provides guidance for designers selecting soft-core architectures for resource-constrained secure systems.